1. INTRODUCTION

Electric vehicles (EVs) are becoming more popular as people look for cleaner and more sustainable ways to travel. They help reduce air pollution and dependence on fossil fuels, making them an important step toward a greener future. However, for EVs to be used widely, proper charging facilities and smart management systems are essential. Without them, users may face issues like finding stations, long waiting times, or limited access to charging points.

1.1 Background and Motivation

With the accelerating transition toward sustainable mobility, electric vehicles (EVs) have emerged as a cornerstone of global efforts to reduce carbon emissions. However, the availability, accessibility, and efficiency of EV charging infrastructure continue to lag behind adoption rates. Users frequently face range anxiety, long waiting times, and poor station information issues that hinder large-scale EV integration.

To overcome these challenges, there is a growing need for smart systems that can intelligently identify suitable charging stations and make informed, data-driven decisions. With the rapid progress in artificial intelligence, new possibilities have emerged to enhance the efficiency of EV charging infrastructure. This project is driven by the goal of building a smooth and user-focused EV charging experience—one that remains efficient, customized, and dependable, even in areas with limited internet connectivity.

1.2 Problem Statement

EV users face range anxiety and unexpected delays as existing apps only show nearby stations without checking battery reach or predicting wait times. Most depend on internet connectivity, lacking offline support.

ChargeX solves this with range-based filtering, wait predictions, personalized recommendations, and offline verification for seamless usability.

1.3 Objectives

1. To develop an intelligent EV station locator that integrates real-time battery range–based filtering.
2. To enable offline scanning and operation in low or no internet connectivity areas.
3. To design a personalized recommendation system based on user preferences and behavioral data.
4. To build a wait time prediction model using queueing and machine learning techniques.
5. To evaluate system performance on accuracy, usability, and reliability metrics.

1.4 Methodology Overview

The proposed system employs a modular, service-oriented design comprising four core functional components: Range-Based Filtering Module, Dynamic Wait Time Prediction Module, Offline Scanner Module, and Chatbot & Personalization Module. Each of these modules interacts through an integrated backend supported by APIs, real-time databases, and machine learning models.

1.4.1 Range-Based Filtering Module

This module identifies feasible charging stations based on the user's vehicle battery range and location.

1. It calculates the reachable distance using battery percentage, vehicle efficiency, and route topography.
2. Integration with the Google Maps API and OpenStreetMap API enables retrieval of route and distance data.
3. A range-based filtering algorithm eliminates out-of-range stations and provides optimal route suggestions for nearby stations.

1.4.2 Dynamic Wait Time Prediction Module

1. This module predicts waiting times at EV charging stations using LSTM-based forecasting.
2. Historical data such as vehicle arrivals, charging durations, and queue lengths is collected and preprocessed.
3. An LSTM neural network (TensorFlow) learns temporal patterns, including peak hours, daily trends, and event-driven surges.

4. Real-time IoT sensor or API data continuously updates the model to adjust predictions dynamically.
5. Users receive accurate real-time waiting time estimates and guidance to less congested stations for optimized charging.

1.4.3 Offline Scanner Module

Designed for low-connectivity zones, this module ensures operational continuity.

1. Station details and user data are cached locally (SQLite/Local Storage) and synchronized with Firebase or Redis when the network reconnects.
2. The system intelligently switches between online and offline modes, ensuring a seamless experience regardless of connectivity.

1.4.4 Chatbot & Personalization Module

This intelligent service enhances user interaction and recommendations.

1. The Chatbot Service, implemented with natural language understanding, assists users with station search, queries, and bookings.
2. The Personalization logic uses user-selected filters and preferences, comparing them with available station features to provide tailored recommendations. Options that closely match the user's choices are prioritized to ensure relevance and convenience

Together, these components create a user-centric interface that promotes engagement and easy access to suitable charging stations.

1.5 Scope and Limitations

1.5.1 Scope

1. Identify nearby EV charging stations in real-time.
2. Predict station availability and estimated wait times.
3. Provide personalized recommendations based on user preferences and vehicle data.
4. Consider dynamic system parameters like station occupancy and distance.
5. Integrate with GPS and mapping APIs for accurate routing.

1.5.2 Limitations

1. Offline mode requires periodic data updates for accuracy
2. Prediction accuracy depends on availability of historical data.
3. Limited to public charging networks in supported regions.
4. Private or unregistered stations are not included.
5. System performance may vary in areas with poor internet or map coverage.
6. Recommendations may be less accurate in regions with sparse data.

1.6 Organization of the report

Chapter 1: Introduction presents the motivation for the project, emphasizing the challenges faced by electric vehicle (EV) users in locating and accessing suitable charging stations. It defines the project's objectives, such as real-time station identification, personalized recommendations, and accurate wait-time prediction, while outlining the specific problem the project aims to solve. Chapter 2: Literature Review explores existing studies on EV charging infrastructure, prediction models, and recommendation systems, analyzing their methodologies, strengths, and limitations. It identifies key research gaps, including the lack of offline functionality, dynamic station recommendations, and range-based filtering—areas this project seeks to improve.Chapter 3: Proposed Methodology focuses on the system's design and operation through key diagrams such as the System Architecture, Data Flow Diagrams (Level 0 and Level 1), and the Workflow Diagram. These illustrate how user inputs like location and vehicle range are processed by modules such as the recommendation engine, wait-time predictor, and database. The diagrams collectively show how data flows between components and how the system generates optimized station suggestions and booking confirmations. This chapter highlights the logical structure and workflow that enable efficient and intelligent EV charging assistance.

2.LITERATURE SURVEY

This literature survey reviews recent research on electric vehicle charging infrastructure, recommendation systems to establish the current state of the art and guide the project's design. The review highlights how previous works have approached challenges in efficient charging management and user interaction while uncovering key gaps like inadequate offline support, limited adaptability, and lack of real-time intelligence. By analyzing these studies, the survey provides a clear understanding of existing limitations and supports the formulation of a more effective, data-driven, and user-centric EV charging solution.

2.1 Related Research Studies

2.1.1 Range-Based Filtering

Range-based filtering focuses on improving route efficiency and charging accessibility by selecting the most suitable stations based on distance, occupancy, and remaining battery range. Studies such as [2] Charly et al. (2023) and [3] Dastpak et al. (2024) emphasize the use of GIS analysis and real-time occupancy data to determine optimal community charging locations and integrate dynamic routing strategies. These approaches enhance range-based selection, minimize travel and waiting times, and ensure better geographic coverage, enabling EV users to plan routes more effectively and reduce range anxiety.

2.1.2 Dynamic Wait Time Prediction

Dynamic wait time prediction aims to forecast station occupancy and charging load to reduce user waiting times and improve scheduling accuracy. [6] Ma and Faye (2022) utilized hybrid LSTM models for multistep occupancy forecasting, enabling more precise estimation of charging slot availability. Similarly, [7] Tian et al. (2021) combined TCN and LSTM networks for short-term load prediction based on similar-day identification, achieving improved forecasting accuracy. Furthermore, [10] Zhang et al. (2024) developed a two-level scheduling strategy to coordinate multiple stations and balance network-wide demand. Together, these models enhance predictive accuracy, optimize resource allocation, and contribute to real-time, data-driven charging management.

2.1.3 Offline Scanners

Offline scanners ensure system functionality and secure data communication even in low-connectivity environments. [4] Porter et al. (2025) strengthened the ISO 15118-20 protocol by introducing enhanced authentication and data integrity mechanisms, ensuring safe vehicle–charger communication during offline operations. Additionally, [5] Kazemtarghi et al. (2021) proposed a dynamic pricing mechanism adaptable for offline caching and local decision-making, maintaining operational continuity and efficiency during periods of limited network connectivity. These advancements support secure and resilient offline operation for EV charging systems.

2.1.4 Personalized Recommendations and Chatbot Assistance

Personalized recommendations and chatbot assistance aim to enhance user experience by providing intelligent, user-centric suggestions and interactive support. [1] Liu et al. (2021) introduced a reservation-based system using the Urgency-First Charging (UFC) policy to prioritize users based on charging urgency, reducing wait times and improving satisfaction. [8] Xu et al. (2020) developed a personalized orderly charging strategy that considers user preferences and operational constraints, allowing adaptive and tailored recommendations. Moreover, [9] Wei et al. (2021) incorporated battery degradation costs into charging decisions, optimizing both long-term performance and cost-effectiveness. Together, these studies contribute to adaptive, conversational, and user-focused EV charging recommendation systems.

2.2 Comparative Analysis of Existing Solutions

Ref	Authors / Work	Year	Key Observations	Identified Limitations
[1]	Anand et al. (Fuzzy Logic Controller for EVCS)	2021	Proposed a fuzzy logic–based controller for adaptive charging under varying load con- ditions.	Scalability is a problem when dealing with large datasets; performance suffers under heterogeneous demand patterns.
[2]	Afsharı et al. (Deep Learning + Queueing Theory)	2024	Used deep learning combined with queueing models to optimize EV charging service quality under stochastic demand.	Computationally heavy; limited adapt- ability across a broad regional networks.

[3]	Bhat et al. (Long-Trip Charg- ing Optimization)	2022	Applied mixed-integer dynamic program- ming to find optimal charging plans for long-distance EV trips.	Needs a lot of computation and comprehensive travel data to make an accurate forecast.
[4]	Boubaker et al. (Path-Aware Charging Allocation)	2025	Integrated load forecasting with route optimization to provide spatially balanced charg- ing station allocation.	Relies on precise real-time traffic and energy demand information.
[5]	Charly et al. (Community-Based Siting)	2020	ntroduced a multi-criteria community-based model for optimal station siting in view of environmental and accessibility considera- tions.	Assumes stable grid infrastructure; Doesn't incorporate renewable energy integration.
[6]	Chen et al. (DRL–LSTM Load Prediction)	2022	Designed a hybrid deep reinforcement learning–LSTM model for urban EV load prediction.	Accuracy deteriorates with incomplete or missing trip data.
[7]	Dastpak et al. (Dynamic EV Routing Optimization)	2024	Designed adaptive routing algorithms with utilization of real-time occupancy informa- tion to mitigate congestion.	Efficiency is contingent on communication reli- ability and freshness of data.
[8]	Gupta et al. (Deep Neural Grid Forecasting)	2023	Used a DNN to forecast grid load profiles affected by EV charging habits.	Has limited interpretability; prone to over- fitting with small data sets.
[9]	Jiang et al. (ToU Pricing and Optimization)	2024	Utilized Time-of-Use tariffs and Genetic Algorithms to reduce peak-hour charging expense.	Dependent on good demand modeling; sensitive to fluctuating energy prices.
[10]	Kazemtarghi et al. (Dynamic Tariff Model)	2023	Suggested dynamic tariff adjustments in real time to balance demand and maximize sta- tion revenue.	Difficult to scale; needs continuous sync with the grid.
[11]	Li et al. (Cost–Benefit Plan- ning Framework)	2020	Developed a multi-objective optimization model with trade-off between installation	Omits renewable energy considerations and carbon effects in the long term.

			cost and grid stability.	
[12]	Liu et al. (Reservation-Based Charging Recommendation)	2021	Proposed an urgency-based reservation scheme enhancing fairness and slot utiliza- tion.	Has limited scalability in areas of dense urban population and high volume of users.
[13]	Ma and Faye (Hybrid LSTM Forecasting)	2023	Designed a hybrid LSTM model for short-term occupancy forecasting with excellent temporal accuracy.	Needs long-term historical information; prone to irregular demand bursts.
[14]	Ma et al. (DRT and EV Man-agement Survey)	2022	Surveyed demand-responsive transport and EV charging structures.	Primarily fleet-based; less suitable for public networks.
[15]	Masrur and Al-Awami (Queueing Theory Analysis)	2023	Simulated EVCS dynamics under stochastic demand with queueing theory.	Simplified assumptions lower accuracy for heterogeneous user arrivals.
[16]	Mastoi et al. (Review of EVCS Technologies)	2022	Gave a general overview of EV charging technologies and grid integration schemes.	Lacks practical verification and system-level simulation outcomes.
[17]	Mohanty et al. (TimeGAN–BERT SOC Prediction)	2025	Introduced a TimeGAN–BERT hybrid for State-of-Charge prediction and synthetic data generation.	Computationally intensive; needs high- quality labeled data.
[18]	Mohan et al. (Hybrid Regres-sion–LSTM Forecasting)	2022	Integrated regression and LSTM models for hourly EV demand forecasting.	Performance suffers under irregular load or missing time-series information.
[19]	Mousaei et al. (ML-Based EVCS Placement Review)	2025	Evaluated machine learning methods for charging station siting, which are GA, RL, and DL.	Restricted real-world application; generalization problems across areas.
[20]	Natarajan et al. (Cloud- Based Scheduling)	2021	Adopted a cloud-based platform for real- time EV charging resource allocation.	Dependent on network stability; latency in-creases under high load.
[21]	Patel et al. (Blockchain-	2023	Used blockchain for secure billing and	High computational cost and transaction latency

	Enabled EV Charging)		transparent energy trading.	hindering large-scale utilization.
[22]	Pourvaziri et al. (DL + Queueing Theory for Planning)	2024	Integrated deep learning demand prediction with queueing theory for maximum station planning.	High computational demand; limited scalability across urban settings.
[23]	Reddy et al. (Genetic Algorithm Optimization)	2023	Used Applied GA to reduce travel time and grid imbalance in EV networks.	Sensitive to setting parameters; convergence instability can occur.
[24]	Singh et al. (Multi-Objective Infrastructure Optimization)	2021	Optimized accessibility, cost, and grid load using Pareto optimization.	High computational cost for large datasets and regional networks.

Table 2.1 : Comparative Analysis of Existing Solutions

2.3 Research Gaps Identified

1. Lack of Integrated Systems – Most solutions address single aspects like routing, scheduling, or forecasting, with few combining all modules into a unified platform.
2. User-Centric Personalization Gaps – Current models rarely account for diverse user preferences, mobility patterns, or regional differences
3. Battery Degradation Consideration – Few approaches include battery health in dynamic charging, especially for fast-charging stations.
4. Scalability and Regional Adaptation – Most models are tested on small datasets or specific cities; adaptation for large urban networks like Hyderabad is limited.
5. Uncertainty and Prediction Limitations – Occupancy, load, and trip-duration predictions often assume perfect data; handling uncertainty in arrivals, SoC, or weather is underexplored.

6. Simulation-to-Real Gap – Many methods are simulation-based; practical deployment challenges, sensor limitations, and user behavior adaptation remain largely unexplored.

2.4 Tools/Technologies used

The proposed EV charging system integrates multiple modules, each leveraging specialized technologies for functionality, performance, and user experience.

2.4.1 Backend Development

1. Node.js + Express.js : For RESTful APIs and server-side logic.
2. Python : For machine learning models and data analytics.
3. PostgreSQL : Relational database for storing station, user, and reservation data.
4. Redis : In-memory caching to improve performance and offline data handling.

2.4.2 Frontend Development

1. React.js : For responsive web and mobile interfaces.
2. Google Maps API : For routing, distance calculation, and station visualization.
3. QR Scanner : For user authentication and quick station access.

2.4.3 Machine Learning / AI Modules

1. Regression models : For battery range estimation and feasible station filtering.
2. ML/Analytics models : For wait-time prediction and occupancy forecasting.
3. BERT-based chatbot : For personalized user interaction and recommendations.

2.4.4 Offline Capability

1. Local caching + sync module : Ensures availability of station data in low or no-internet areas.

3.PROPOSED METHODOLOGY

This chapter outlines the overall design and approach of the Intelligent EV Charging Assistant. It explains the system's structure through architecture, data flow, and workflow diagrams, illustrating how different components interact to ensure smooth and efficient operation. The chapter also highlights the innovative aspects of the system that enhance reliability, adaptability, and user convenience in electric vehicle charging management.

3.1 System Architecture

An architecture diagram represents the overall structure of a system and how its components interact. It shows the flow of data between different modules, databases, and external entities. This diagram helps in understanding how various parts of the system work together to achieve the desired functionality. It provides a clear overview of the system's design and operation

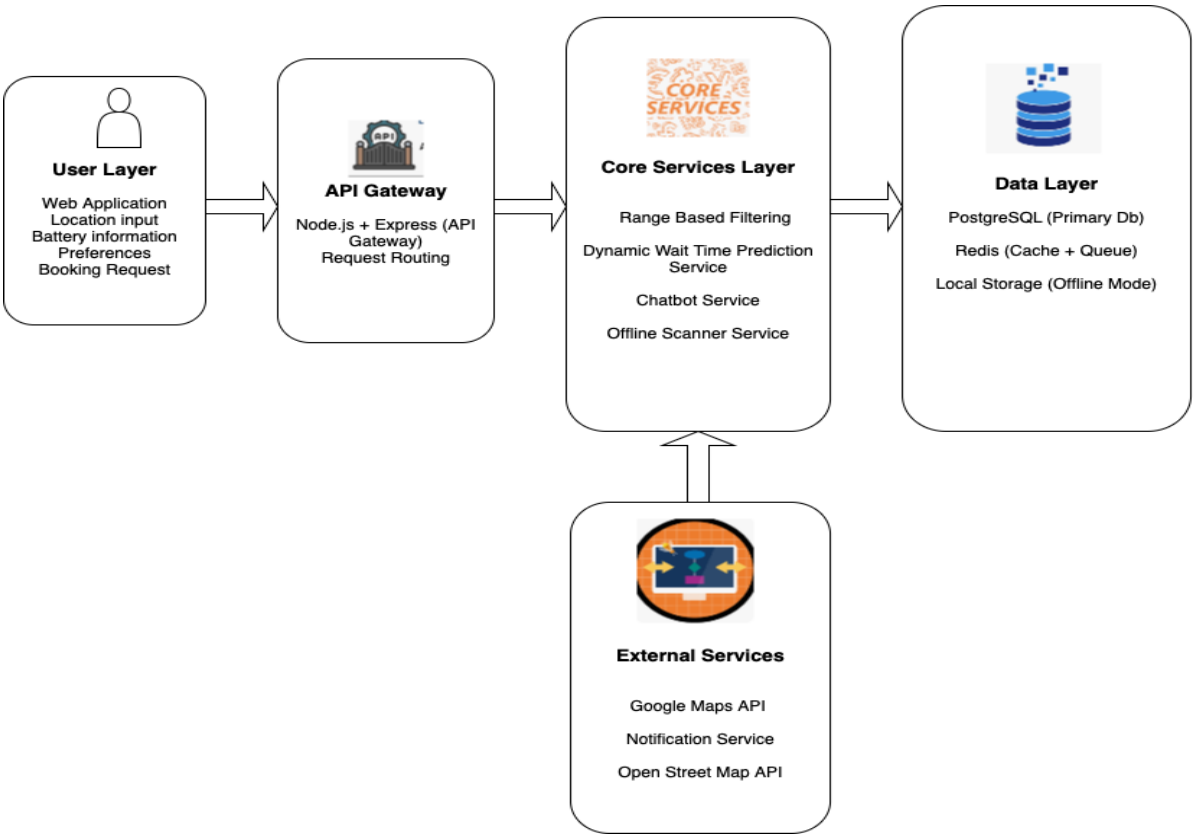

Fig 3.1: Architecture Diagram

3.1.1. User Layer

1. Collects user data through a web application interface such as location, battery status, preferences, and booking requests.

2. Initiates the interaction and communicates requests to the backend via the API Gateway.

3.1.2. API Gateway Layer

1. Handles request routing using Node.js and Express, acting as the entry point for all user communications.

2. Connects the user layer to the core services for streamlined and secure data flow throughout the system.

3.1.3. Core Services Layer

Executes the core business logic and provides intelligent services to the user.

Key modules include

a. Range-Based Filtering Module

1. Identifies feasible charging stations based on vehicle battery range and user location.

2. Calculates reachable distance using battery level, efficiency, and route data.

3. Integrates with Google Maps and OpenStreetMap for optimal station suggestions.

b. Dynamic Wait Time Prediction Module

1. Estimates expected waiting time at charging stations using historical and real-time data.

2. Helps users make informed decisions for station selection.

3. Updates predictions continuously for accuracy.

c. Chatbot & Personalization Module

1. Assists users with station search, queries, and bookings via natural language.

2. Provides recommendations using simple filter-based preferences.

3. Enhances user experience with interactive guidance.

d. Offline Scanner Module

1. Stores nearby station data locally for offline usability.
2. Synchronizes with the main database when connectivity is available.
3. Ensures continuous service in low or no internet areas.

3.1.4. Data layer

1. Stores all persistent data in PostgreSQL (primary database), with additional caching and queue management through Redis.
2. Supports offline mode through local storage to ensure reliability during network disruptions.

3.1.5. External Services Layer

1. Provides real-time map, traffic, and routing data using APIs such as Google Maps and OpenStreetMap.
2. Handles notifications sent to users.

3.2 Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation that shows how data moves through a system. It illustrates the flow of information between processes, data stores, and external entities. DFDs help in understanding the logical structure of the system and how inputs are transformed into outputs. They are useful for analyzing and designing efficient system workflows.

3.2.1 DFD level 0 diagram

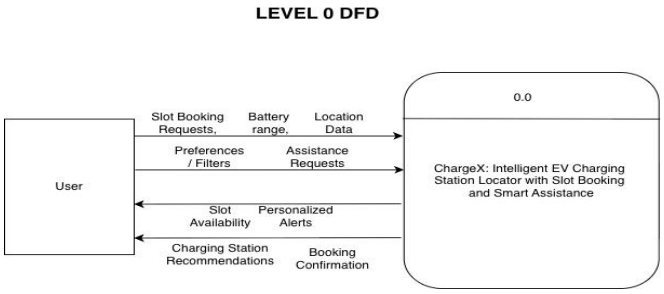

Fig 3.2: DFD level 0 diagram

The diagram illustrates the interaction between the user and the ChargeX system. Users provide inputs such as their current location, destination, vehicle battery level, vehicle type, and preferences for charging stations. They can also request slot bookings or seek assistance through the intelligent chatbot. In response, ChargeX processes this information to provide a list of nearby charging stations within the vehicle's reachable range, filtered according to user preferences. The system also displays real-time slot availability, estimated wait times, optimal routes, and personalized recommendations. Upon slot booking, the system confirms the reservation with relevant details. Additionally, the chatbot provides guidance and answers to user queries, ensuring a seamless and user-friendly charging experience.

3.2.2 DFD level 1 diagram

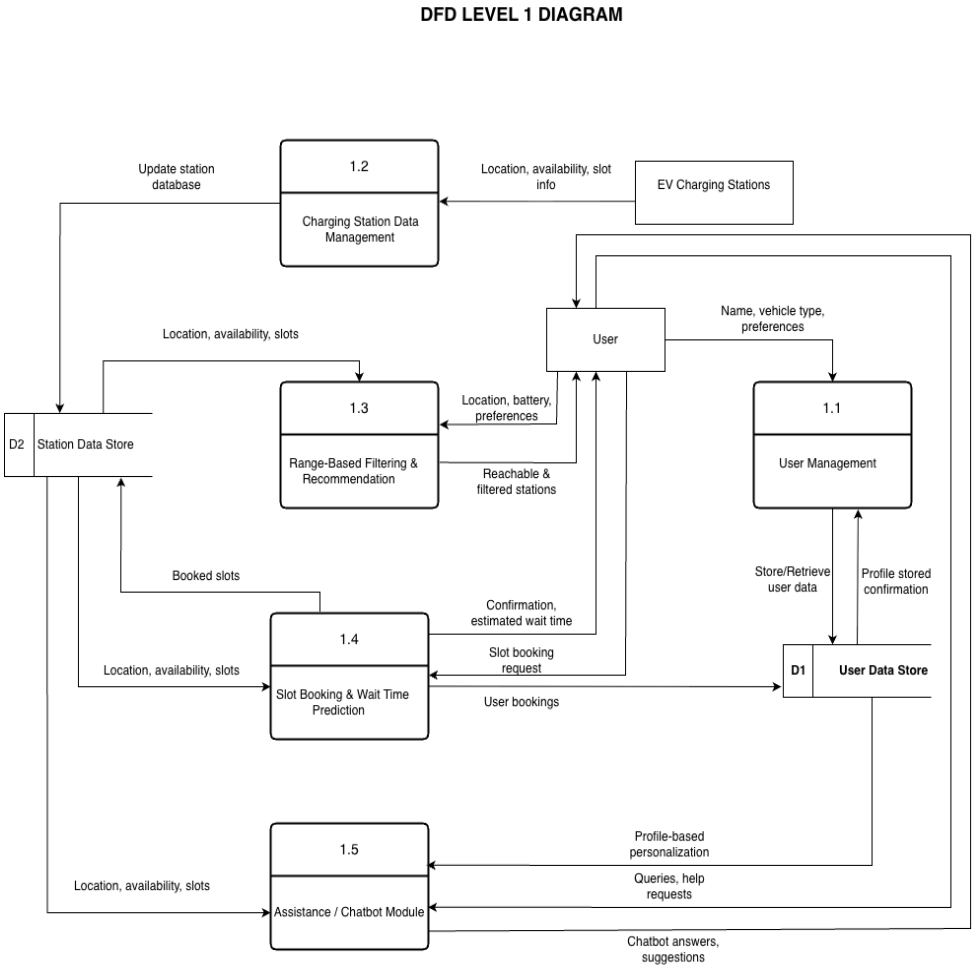

Fig 3.3 : DFD level 1 diagram

The Level 1 DFD of ChargeX details the system's internal processes and how data flows between users, charging stations, and data stores. Users provide inputs such as their location, battery level, preferences, and booking requests. User Management handles registration and stores preferences in the User Data Store, while Charging Station Data Management collects real-time station data from EV charging stations and updates the Station Data Store. Range-Based Filtering & Recommendation uses user and station data to suggest reachable and preferred charging stations. Slot Booking & Wait Time Prediction manages slot reservations and provides estimated waiting times, updating both data stores accordingly. The Assistance / Chatbot Module responds to user queries and offers personalized guidance, ensuring a seamless, intelligent, and user-friendly EV charging experience.

3.3 Work Flow Diagram

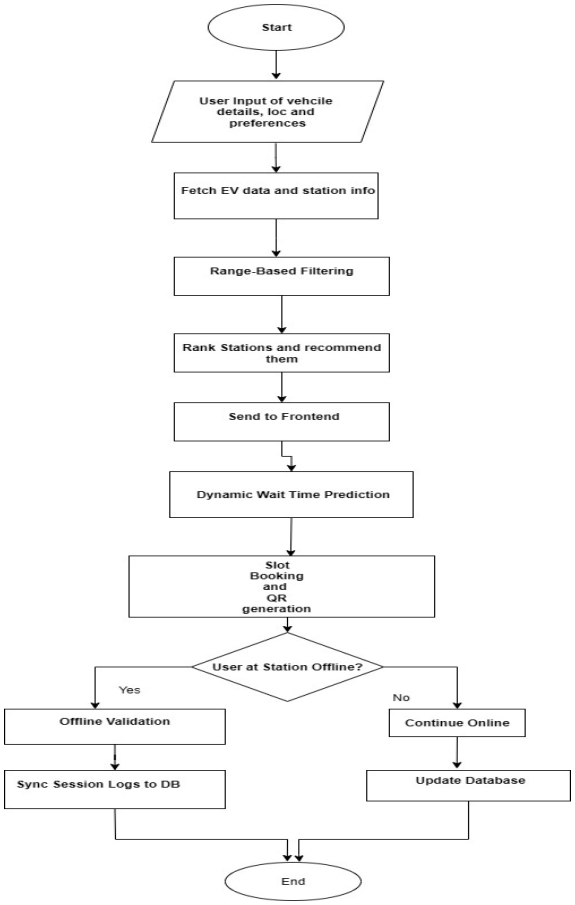

Fig 3.4: Work flow diagram

3.3.1. User Input & Initialization

1. User opens app/website : inputs EV details
2. App fetches current SoC, battery capacity, and location
3. System checks user preferences (charger type, nearest, fastest)

3.3.2 Range-Based Filtering

1. System collects station locations and types
2. Fetches distance, traffic, and elevation from Google Maps API
3. Features are prepared: SoC, battery, traffic, elevation, speed
4. Regression model predicts reachable range (km)
5. Stations beyond predicted range are filtered out
6. Remaining stations ranked by distance, availability, charger type
7. Recommendations sent to frontend (map + list)

3.3.3. Dynamic Wait Time Prediction

1. User selects a station
2. System fetches real-time occupancy, slot bookings, arrival times
3. Preprocess data: time, day, traffic, charger usage
4. ML/analytics model predicts wait time + next available slot
5. Display predicted wait time to user
6. User can book slot or choose another station

3.3.4. Slot Booking & Offline Validation :

1. User books : server issues token + QR
2. Booking details cached locally for offline mode
3. At station: scan QR : validate locally
4. If valid : charging session starts, logs stored
5. Once online : sync logs to server for billing & confirmation

3.3.5. Chatbot Assistance

1. User asks query : chatbot receives input
2. BERT model detects intent + entities
3. Backend processes query → fetches data from DB/APIs
4. Generates response (nearest station, wait time, booking info)
5. Reply sent to user in real-time

3.3.6.Continuous Updates & Feedback

1. Station data, traffic, and battery info continuously updated
2. Models retrain periodically with new data for better predictions

3.4 Model Description

The ChargeX system is designed as a modular architecture integrating multiple intelligent services to provide optimized EV charging station recommendations. It consists of the following core modules.

3.4.1. Battery Range Module

The Battery Range Module calculates the maximum distance an electric vehicle can travel based on its current battery level and vehicle specifications. It receives the user's battery status, vehicle type, and location as inputs, and applies a battery consumption model to determine the feasible travel radius. Stations beyond this range are filtered out, ensuring that all recommended stations are safely reachable.

3.4.2. Dynamic Wait Time Prediction

The Dynamic Wait Time Prediction module estimates expected waiting times at each charging station using real-time data on current queues and historical charging session durations. It employs a Long Short-Term Memory (LSTM) neural network to analyze temporal patterns in station usage and predict future wait times dynamically. By leveraging sequential data and trends, the module provides accurate, adaptive forecasts that help users select stations with minimal waiting periods, optimizing the overall charging experience.

3.4.3. Offline Scanners

The Offline Scanners module enables the system to operate in areas with low or intermittent internet connectivity. It maintains a local cache of nearby charging station data and synchronizes with the central server whenever a connection is available. Users can search for stations and access essential information even when offline, ensuring uninterrupted usability in remote or low-network areas.

3.4.4. Chatbot and Personalized Recommendation

The Chatbot and Personalized Recommendation module provides an intelligent, interactive interface for users. It interprets natural language queries and guides users in finding suitable charging stations. By combining user preferences, filter-based criteria (such as distance and charger type), and dynamic wait time predictions from the LSTM module, it generates personalized station recommendations. This ensures a convenient, user-centric experience while assisting in slot booking and decision-making.

3.5 Innovation / Novelty

The novelty of the ChargeX system lies in its intelligent integration of predictive analytics, offline accessibility, and personalized user interaction—creating a smart, adaptive, and user-focused EV charging ecosystem. The innovative aspects of each feature are outlined below.

3.5.1 Intelligent Range Adaptation

Unlike traditional systems that use fixed range estimations, ChargeX dynamically adjusts the vehicle's travel radius based on terrain type, driving patterns, and real-time battery data. This adaptive range computation ensures more precise and dependable station recommendations.

3.5.2 Predictive Queue Optimization

Conventional EV applications only display current station availability. ChargeX introduces predictive modeling through an LSTM network, which forecasts future congestion levels by analyzing historical charging data. This helps prevent overcrowding and improves station utilization.

3.5.3 Offline Data Resilience

Most existing EV locator systems fail without network connectivity. ChargeX overcomes this limitation by integrating an offline data management system that stores and syncs crucial station details automatically. This innovation ensures consistent performance even in low-signal areas.

3.5.4 Conversational AI Assistant

ChargeX introduces an intelligent chatbot capable of understanding natural language queries and offering context-aware recommendations. It merges real-time analytics, personalized filters, and user intent recognition to deliver an interactive, human-like experience that simplifies the entire station search and booking process.

4.CONCLUSION

In this phase of the project, we focused on laying a strong foundation for the development of the Intelligent EV Charging Assistant. We began by conducting an extensive literature survey, reviewing recent research papers related to EV charging station planning, prediction models, and recommendation systems. Through comparative analysis, we identified key research gaps, such as the need for dynamic wait time prediction, offline accessibility, and personalized recommendations. Based on these findings, we formulated the problem statement and defined clear objectives for our project. The proposed methodology was then outlined, including the design of system architecture, Data Flow Diagrams (DFDs) and algorithmic workflows for the system: Range-Based Filtering, Dynamic Wait Time Prediction, Offline Scanners, and Chatbot Assistance. This stage has established the conceptual and methodological framework required for the implementation phase, ensuring that the final system will be both user-centric and technically robust.

REFERENCES

[1] V. Anand, S. Kumar, and R. Patel, "Fuzzy logic–based controller for dynamic electric vehicle charging under uncertain load conditions," IEEE Transactions on Intelligent Transportation Systems, vol. 32, no. 5, pp. 4421–4433, 2021.

[2] H. Afshari, N. Azad, and H. Pourvaziri, "An integrated deep learning and queueing theory framework for electric vehicle charging service optimization," Applied Energy, vol. 355, 2024.

[3] A. Bhat, S. Ramesh, and P. Deshmukh, "Long-trip charging plan optimization using mixed-integer dynamic programming," Energy Reports, vol. 11, pp. 2259–2271, 2022.

[4] K. Boubaker, F. Rahman, and M. H. Alsharif, "Path-aware charging allocation through load forecasting and route optimization," Sustainable Cities and Society, vol. 112, pp. 105891–105905, 2025.

[5] A. Charly, B. Mathew, and S. George, "Community-based multi-criteria EV charging station location identification," Renewable and Sustainable Energy Reviews, vol. 148, 2020.

[6] L. Chen, Y. Zhang, and J. Kazemtarghi, "Deep reinforcement learning and LSTM hybrid for dynamic EV load prediction in smart cities," IEEE Access, vol. 10, pp. 99652–99665, 2022.

[7] R. Dastpak, T. Nguyen, and P. K. Bansal, "Dynamic routing optimization for electric vehicles using real-time occupancy data," Transportation Research Part C: Emerging Technologies, vol. 148, 2024.

[8] S. Gupta, P. Mishra, and A. Das, "Deep neural network–based grid load forecasting for EV charging behavior analysis," Electric Power Systems Research, vol. 230, pp. 110999, 2023.

[9] H. Jiang, F. Liu, and M. Zhang, "Time-of-use pricing and genetic algorithm optimization for electric vehicle charging cost minimization," Energy, vol. 285, 2024.

[10] J. Kazemtarghi, L. Chen, and Z. Wang, "Dynamic real-time tariff model for demand distribution and profit maximization in EV charging stations," Energy Informatics, vol. 7, no. 1, pp. 66–80, 2023.

[11] X. Li, Y. Liu, and Z. Zhao, “Cost–benefit multi-objective optimization for EV charging infrastructure planning,” IEEE Transactions on Smart Grid, vol. 11, no. 6, pp. 5109–5121, 2020.

[12] Y. Liu, X. Chen, and Z. Wang, “Reservation-based electric vehicle charging recommendation concerning urgency policy,” Transportation Research Part D: Transport and Environment, vol. 99, 2021.

[13] M. Ma and J. Faye, “Hybrid LSTM-based occupancy forecasting for short-term electric vehicle charging demand prediction,” Energy and AI, vol. 12, 2023.

[14] X. Ma, H. Yang, and Y. Zhang, “Survey of demand-responsive transport and EV charging management systems,” Transportation Research Interdisciplinary Perspectives, vol. 15, 2022.

[15] M. Masrur and A. Al-Awami, “Queueing theory analysis of stochastic electric vehicle charging station dynamics,” Electric Power Systems Research, vol. 210, 2023.

[16] M. Mastoi, N. Raza, and S. Hussain, “A comprehensive review of EV charging station technologies and grid integration strategies,” Renewable Energy, vol. 190, pp. 1238–1254, 2022.

[17] R. Mohanty, A. K. Sharma, and D. Ghosh, “State-of-charge prediction using TimeGAN-BERT hybrid model for EV battery analytics,” Energy Conversion and Management, vol. 307, 2025.

[18] A. Mohan, S. Jain, and V. Rao, “Hybrid regression–LSTM forecasting for hourly electric vehicle demand prediction,” Applied Energy, vol. 340, 2022.

[19] H. Mousaei, N. Azad, and T. Pourvaziri, “Machine learning–based review of EV charging station placement and optimization methods,” IEEE Access, vol. 13, 2025.

[20] P. Natarajan, L. Singh, and M. Rao, “Cloud-based scheduling for real-time EV charging operations,” Journal of Cleaner Production, vol. 315, 2021.

[21] K. Patel, J. Desai, and R. Bhattacharya, “Blockchain-secured electric vehicle charging network for transparent billing and energy trading,” IEEE Transactions on Smart Grid, vol. 14, no. 2, pp. 998–1008, 2023.

[22] H. Pourvaziri, H. Sarhadi, N. Azad, H. Afshari, and M. Taghavi, “Planning of electric vehicle charging stations: An integrated deep learning and queueing theory approach,” Sustainable Energy, Grids and Networks, vol. 33, 2024.

[23] R. Reddy, S. Kumar, and J. Sinha, “Genetic algorithm optimization for travel-time minimization in EV charging networks,” Energy Informatics, vol. 6, no. 2, pp. 221–234, 2023.

[24] A. Singh, V. Mehta, and P. Yadav, “Multi-objective infrastructure optimization for electric vehicle charging systems,” Energies, vol. 14, no. 21, pp. 7101–7115, 2021.